STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-HENSELIZATION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1250-1260 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 1262-1277, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 35 of queue_examples-defos.json (pb=808, stmt 1250-1260, proof 1262-1277).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0DY1: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1031–1040. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0AGU: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1277–1277. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DYE: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1277–1277. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Example 0DY1 let P be a k-algebra. Let $J \subset P$ be an ideal. Denote (P^h, J^h) the henselization of the pair (P, J). There is a natural functor*

$$\mathcal{D}ef_P \longrightarrow \mathcal{D}ef_{P^h}$$

of deformation categories.

Proof. Given a deformation of P we can take the henselization of it to get a deformation of the henselization; this is clear and we encourage the reader to skip the proof. More precisely, let $(A, Q) \to (k, P)$ be a morphism in $\mathcal{F}$, i.e., an object of $\mathcal{D}ef_P$. Denote $J_Q \subset Q$ the inverse image of J in Q. Let (Q^h, J_Q^h) be the henselization of the pair (Q, J_Q). Recall that $Q \to Q^h$ is flat (More on Algebra, Lemma

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0AGU) and hence Q^h is flat over A. By More on Algebra, Lemma 0DYE we see that the map $Q^h \to P^h$ induces an isomorphism $Q^h \otimes_A k = Q^h \otimes_Q P = P^h$. Hence $(A, Q^h) \to (k, P^h)$ is the desired object of $\mathcal{D}ef_{P^h}$. $\square$